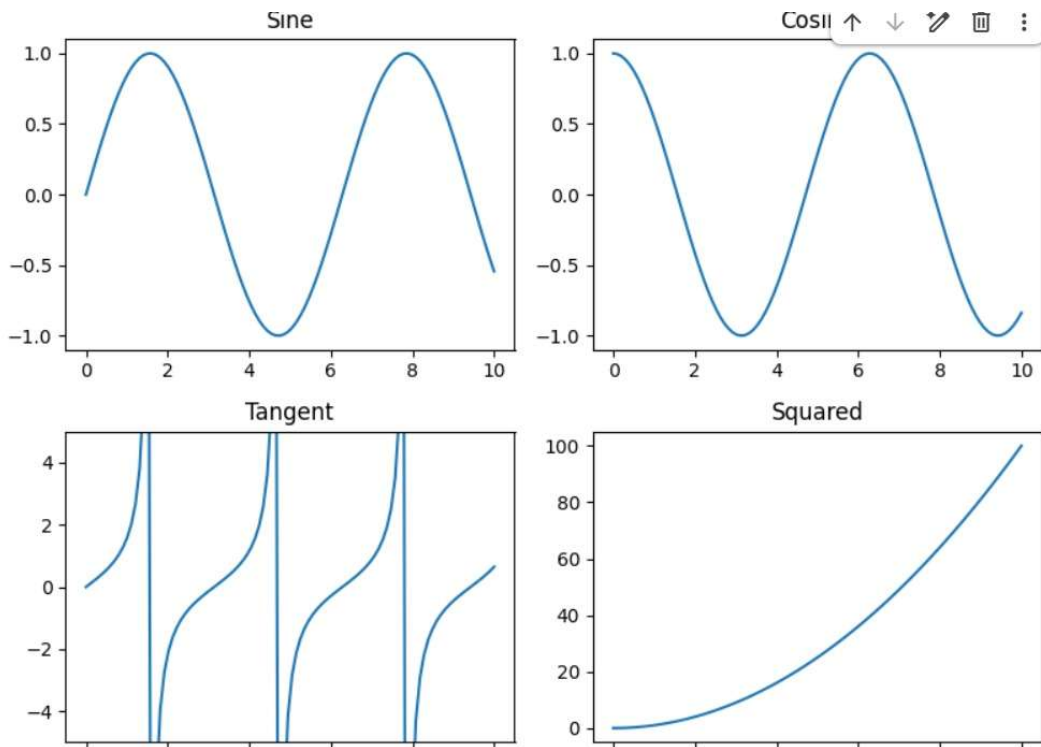




```
import pandas as pd

# Example data
df = pd.DataFrame({
    "Name": ["Alice", "Bob", "Charlie", "David"],
    "Age": [25, 17, 30, 22],
    "City": ["Chennai", "Madurai", "Chennai", "Coimbatore"]
})

# Filter rows where Age is greater than 18
filtered_df = df[df["Age"] > 18]

print(filtered_df)
```

	Name	Age	City
0	Alice	25	Chennai
2	Charlie	30	Chennai
3	David	22	Coimbatore

```

▶ import pandas as pd

# Create sample data
data = {
    "Name": ["Alice", "Bob", "Charlie", "David", "Eva"],
    "Age": [25, 17, 30, 22, 28],
    "City": ["Chennai", "Madurai", "Chennai", "Coimbatore", "Madurai"],
    "Marks": [85, 65, 90, 70, 88]
}

df = pd.DataFrame(data)

# Filter using multiple conditions:
# Age greater than 18 AND Marks greater than 75
filtered_data = df[(df["Age"] > 18) & (df["Marks"] > 75)]

print(filtered_data)

```

```

**      Name  Age   City  Marks
0    Alice   25  Chennai    85
2  Charlie   30  Chennai    90
4      Eva   28  Madurai    88

```

```
▶ import pandas as pd

# Create sample data
data = {
    "Name": ["Alice", "Bob", "Charlie", "David", "Eva"],
    "Age": [25, 17, 30, 22, 28],
    "Marks": [85, 65, 90, 70, 88]
}

df = pd.DataFrame(data)

# Sort by Age in ascending order
sorted_data = df.sort_values(by="Age")

print(sorted_data)
```

```
..      Name  Age  Marks
1      Bob   17    65
3    David   22    70
0    Alice   25    85
4      Eva   28    88
2  Charlie   30    90
```



```
print(df.isnull().sum().sum())
```

... Original Data:

	Name	Age	Marks	City
0	Alice	25.0	85.0	Chennai
1	Bob	NaN	65.0	Madurai
2	Charlie	30.0	NaN	Chennai
3	David	22.0	70.0	NaN
4	Eva	NaN	88.0	Madurai

Missing values (True/False):

	Name	Age	Marks	City
0	False	False	False	False
1	False	True	False	False
2	False	False	True	False
3	False	False	False	True
4	False	True	False	False

Missing values in each column:

```
Name      0
Age        2
Marks      1
City       1
dtype: int64
```

Total missing values:

4



```
import pandas as pd

# Example DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', None, 'David'],
    'Age': [25, None, 30, 28],
    'City': ['Chennai', 'Delhi', 'Mumbai', None]
})

print("Original DataFrame:")
print(df)

# 5. Removing missing values
df_cleaned = df.dropna()

print("\nDataFrame after removing rows with missing values:")
print(df_cleaned)
```

... Original DataFrame:

	Name	Age	City
0	Alice	25.0	Chennai
1	Bob	NaN	Delhi
2	None	30.0	Mumbai
3	David	28.0	None

DataFrame after removing rows with missing values:

	Name	Age	City
0	Alice	25.0	Chennai

```
▶ # Create a sample DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Alice', 'David', 'Bob'],
    'Age': [25, 30, 25, 28, 30],
    'City': ['Chennai', 'Delhi', 'Chennai', 'Mumbai', 'Delhi']
})

print("Original DataFrame:")
print(df)

# 6. Remove duplicate data
df_cleaned = df.drop_duplicates()

print("\nDataFrame after removing duplicates:")
print(df_cleaned)
```

```
... Original DataFrame:
   Name  Age  City
0  Alice   25  Chennai
1   Bob   30   Delhi
2  Alice   25  Chennai
3  David   28  Mumbai
4   Bob   30   Delhi

DataFrame after removing duplicates:
   Name  Age  City
0  Alice   25  Chennai
1   Bob   30   Delhi
3  David   28  Mumbai
```



```
'Department': ['IT', 'HR', 'IT', 'HR'],
'Salary': [50000, 45000, 60000, 55000]
})

print("Original DataFrame:")
print(df)

# 7. Aggregation
print("\nAggregation results:")

print("Total Salary:", df['Salary'].sum())
print("Average Salary:", df['Salary'].mean())
print("Minimum Salary:", df['Salary'].min())
print("Maximum Salary:", df['Salary'].max())
print("Number of Employees:", df['Salary'].count())
```

... Original DataFrame:

	Name	Department	Salary
0	Alice	IT	50000
1	Bob	HR	45000
2	Charlie	IT	60000
3	David	HR	55000

Aggregation results:

Total Salary: 210000

Average Salary: 52500.0

Minimum Salary: 45000

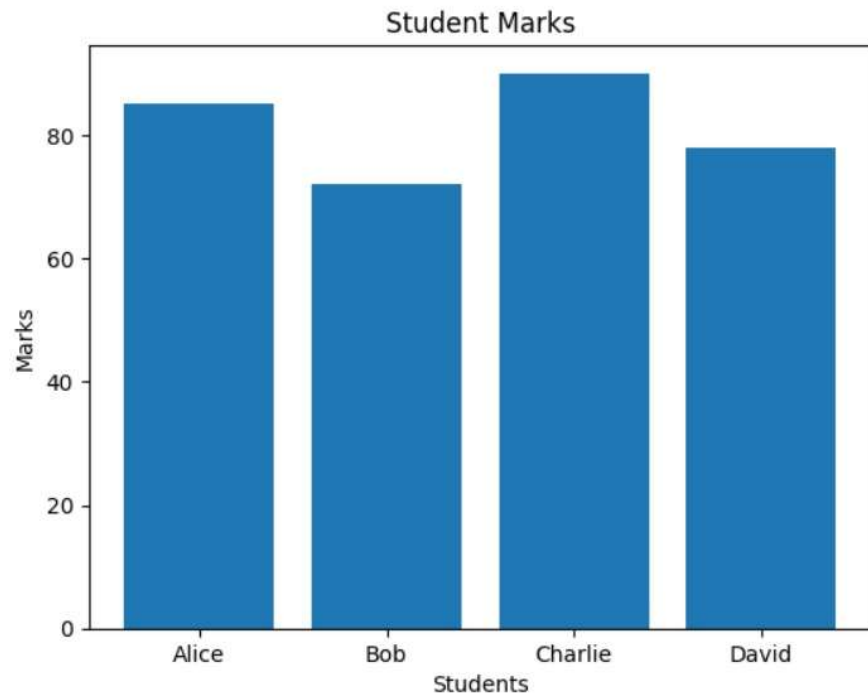
Maximum Salary: 60000

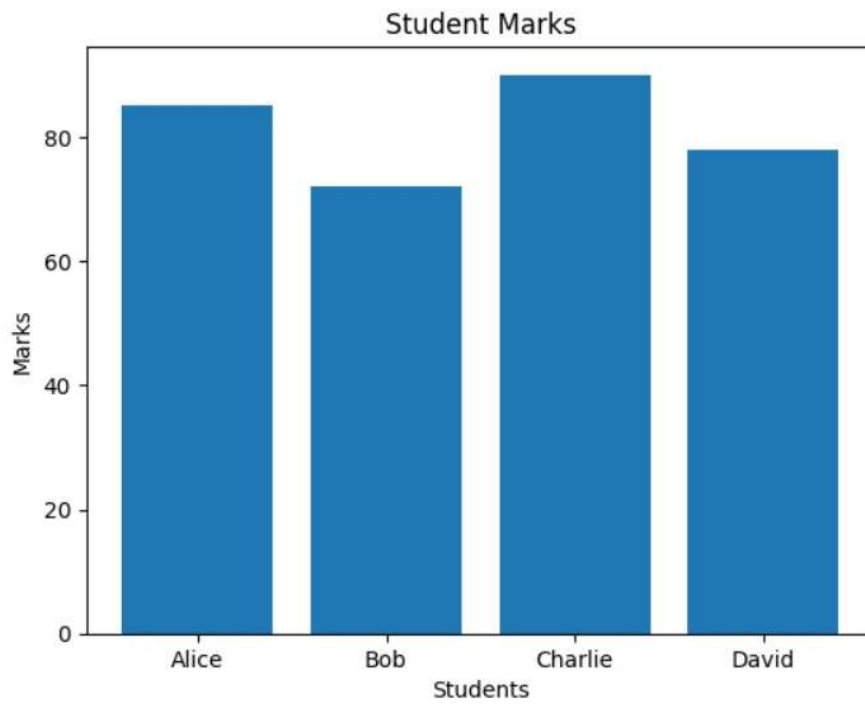
Number of Employees: 4

[8]
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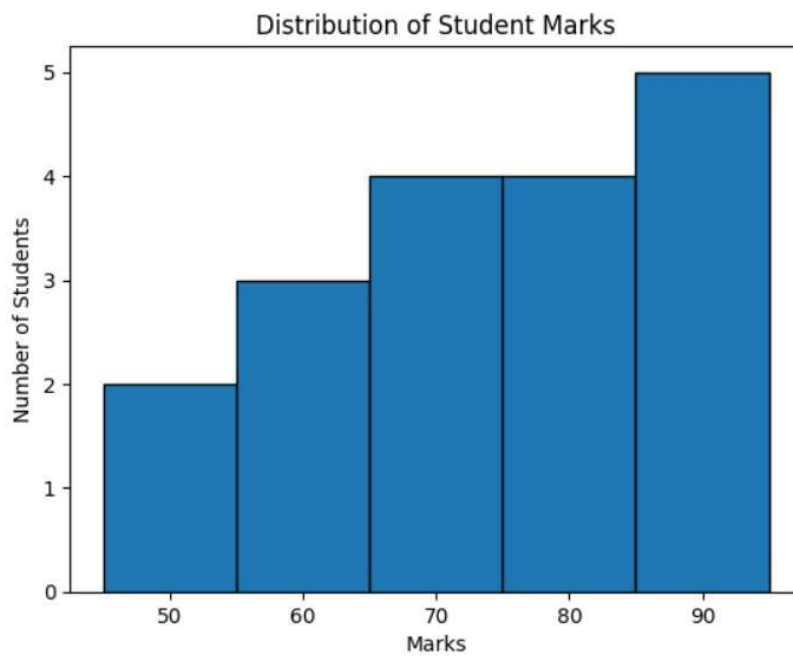
plt.show()

1

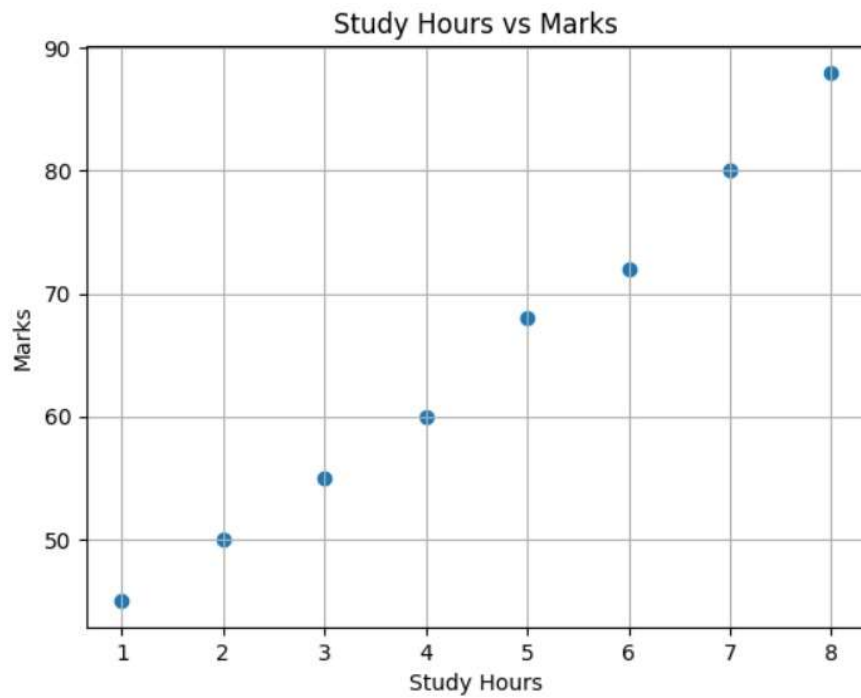




...



10



10

Marks Distribution by Subject

